Are nutrient policy impacts on recreation in Lake Erie as murky as the water?*

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Abstract

Harmful algal blooms have re-emerged as Lake Erie's central management challenge, and the United States and Canada have committed to a policy to reduce them. The policy is likely to affect two very different groups of recreational users in different ways: beachgoers, whose visits depend on water conditions, and anglers, whose visits depend on the lake's fish populations. We designed and administered contingent-behavior surveys of 1,921 beachgoers and 243 anglers that record actual trips alongside the trips respondents would take under scenarios varying bloom severity, water safety, and fish populations. We estimate negative binomial travel-cost models that pool both, so that real behavior anchors the estimates while the scenarios identify responses to conditions the lake has not yet experienced, and we feed Ecopath with Ecosim simulations of the policy into the demand functions. We find that the policy modestly benefits beachgoers but harms anglers through reduced fish abundance, and that per person the anglers' losses are much larger than the beachgoers' gains. In the Western Basin, where the policy shrinks blooms from severe to moderate, beach trips rise by only 1.4%, because beachgoers avoid moderate blooms almost as much as severe ones. Anglers respond mainly to water safety and fish abundance rather than to bloom severity, and the projected decline in fish abundance cuts their trips by 30.5%, an annual loss of about $1,048 per angler, two to twelve times the per-person gain for beachgoers. Judging the policy only by its effect on blooms would therefore overstate its recreational benefits and overlook a group that bears concentrated losses.

Keywords: recreation demand; travel cost method; harmful algal blooms; revealed and stated preference; Lake Erie; nutrient policy; ecosystem modeling.

JEL classification: Q26; Q51; Q53; Q22; Q57.

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1 Introduction

In the late 1960s, national media declared Lake Erie dead. Nutrient pollution had overwhelmed the shallowest, warmest, and most biologically productive of the Great Lakes (U.S. Environmental Protection Agency, 2025), and the images of algae-choked beaches helped spur the creation of the U.S. Environmental Protection Agency, the Clean Water Act of 1972, and, in the same year, the binational Great Lakes Water Quality Agreement (GLWQA), which launched an aggressive phosphorus-reduction program aimed at point sources such as wastewater treatment plants and detergents (De Pinto et al., 1986). By the late 1980s the annual load target of 11,000 metric tons had been met and water quality had improved markedly (Burns et al., 2005)—one of the clearest demonstrations that nutrient control works even at the scale of a Great Lake. Today Lake Erie supports a regional economy of more than $50 billion per year through recreation, fishing, tourism, and shipping (Watson et al., 2016).

The recovery did not last. Over the past two decades the lake has re-eutrophied (Auer et al., 2010; Michalak et al., 2013; Scavia et al., 2014): harmful algal blooms (HABs) have returned to the western basin, and the hypoxic zone—water too low in dissolved oxygen for most aquatic life—has expanded in the central basin, driven by nutrient loads interacting with climate change and invasive species. Policy has responded with the familiar remedy: the GLWQA was updated to require a further 40% reduction in annual phosphorus loads relative to 2008 levels, this time aimed primarily at agricultural runoff (Governments of Canada and the United States, 2012; Environment and Climate Change Canada and U.S. Environmental Protection Agency, 2016).

This time, however, the remedy is not unambiguous. Phosphorus feeds not only the blooms but also the food web that sustains one of the world's great freshwater fisheries, and the International Joint Commission has warned that further reductions could depress offshore productivity and jeopardize the fishery's sustainability (International Joint Commission, 2020). A policy designed around a single objective—fewer blooms—may therefore redistribute value across the lake's users rather than simply create it: better beaches for some and fewer fish for others, the kind of uneven distribution of gains and losses that environmental policies commonly produce (Fullerton, 2011; Hsiang et al., 2019).

This paper quantifies the recreational welfare consequences of the 40% reduction for Lake Erie's two largest user groups—beachgoers and anglers—and asks who wins, who loses, and by how much. The 40% reduction is the policy the two countries have committed to, and it is the focus of our analysis. As a basis for comparison, we also evaluate a deeper 80% reduction; because fish-population changes were simulated only for the 40% reduction, this comparison covers beach recreation alone. We designed and administered two contingent-behavior surveys:

a 2023 survey of 1,921 beachgoers across the four U.S. shoreline states and a 2024 survey of 243 anglers drawn from state fishing-license records. Each survey records respondents' actual day trips alongside the trips they would take under experimentally assigned changes in bloom size, foul odor, harmful bacteria, and—for anglers—fish populations. We estimate joint revealed preference/stated preference (RP/SP) negative binomial travel-cost models, separately by basin for beachgoers, and link the estimated demand functions to Ecopath with Ecosim (EwE; Christensen and Walters, 2004) simulations of the policy produced by the project's ecological team, translating predicted ecological change into predicted trips and dollars. EwE food-web models of Lake Erie have been used to project how the lake's food web responds to invasive species and to nutrient loading and hypoxia (Zhang et al., 2016, 2019, 2023).

Combining stated preference data—hypothetical responses used to estimate ex-ante welfare—with revealed preference data—observed behavior used to estimate ex-post welfare—is a well-established approach in recreation demand analysis (Cameron, 1992; Adamowicz et al., 1994; Whitehead and Lew, 2020), and its contingent-behavior variant, in which respondents report how their trips would change under hypothetical conditions and those responses are pooled with their actual trips, has a long track record in valuing environmental quality (Englin and Cameron, 1996; Whitehead et al., 2000). We build on this tradition, and our design adds three features that the policy question requires. Bloom extent enters as a categorical severity factor tied to NOAA's bloom classification rather than as a continuous per-mile effect, which lets the data reveal the threshold response that drives the Western Basin results: Western Basin beachgoers cut their trips by about the same amount whether a bloom is moderate or severe, so trips rebound only when a bloom shrinks all the way to mild. Basin-specific models estimated on exclusive single-basin visitors give each basin its own travel-cost coefficient, so consumer surplus is not blurred by averaging travel costs across destinations. And the angler survey pairs a bloom experiment with a fish-population experiment on the same respondents, which shows that hypothetical bias is a property of the experiment rather than of the anglers: large in the bloom scenarios and absent in the fish-population scenarios.

The paper's main contributions are in what these demand functions make possible. First, prior Lake Erie recreation studies have focused on particular beaches or on the western basin (Sohngen et al., 1999; Wolf et al., 2019); we estimate demand and welfare separately for the Western, Central, and Eastern basins using respondents from all four U.S. shoreline states, which proves essential because the three basins respond to blooms in qualitatively different ways. Second, we link the estimated demand functions to simulated outputs of a calibrated ecosystem model—Ecopath with Ecosim—for the 40% phosphorus-reduction

policy, translating predicted ecological change into predicted trips and welfare. Earlier work has fed biological model output into recreation demand: Massey et al. (2006) drive an angler site-choice model with a bioeconomic model of a coastal fishery, Lupi et al. (2003) run the predicted fish-population effects of sea lamprey control through a Great Lakes angling model, and Melstrom et al. (2015) use modeled fish biomass as a site attribute; Ready et al. (2018) pool RP and SP data and project expert-elicited invasive-species scenarios through an angler demand model for the Great Lakes. To our knowledge, no study has run the policy simulations of a calibrated ecosystem model through a demand model estimated on pooled RP and SP data. Third, we evaluate the same policy simultaneously for beachgoers and anglers. This matters because of a tradeoff the ecological literature has recently documented. Scavia et al. (2024) combine nearly a century of Lake Erie commercial-harvest records (1928–2022) with hypoxia[1] modeling and conclude that the current phosphorus-load targets appear overly restrictive relative to the needs of the fishery—consistent with the long-recognized dependence of fish production on nutrient supply (Ney, 1996), the documented reshaping of Lake Erie's fish community by earlier nutrient controls (Ludsin et al., 2001), and the IJC's warning above. That literature establishes the tradeoff in ecological and commercial-harvest terms; ours provides the first economic quantification of its recreational-welfare side. Understanding both sides of this tradeoff is directly relevant to the discussions regulators and fishery managers are having about the policy, and the model's outputs are designed to support decision makers who need to know how the intervention propagates through the interconnected ecological and economic systems.

Three findings stand out. First, harmful bacteria—the health-relevant bloom attribute—are the strongest deterrent for both groups, cutting beachgoer trips by roughly a third in every basin and standing as the only bloom characteristic that significantly reduces angler trips; foul odor, a nuisance rather than a health threat, matters consistently but less, and for anglers hardly at all. Second, the response to bloom extent is sharply basin-specific: proportional in the Eastern Basin, absent in the Central Basin, and threshold-like in the Western Basin, where Moderate and Severe blooms deter visitors almost identically. The threshold has direct policy content: because the 40% reduction moves the Western Basin bloom only from Severe to Moderate, it raises beach trips by just 1.4%; the deeper 80% reduction, which restores Mild background conditions, raises them by 20.1%.

Third—and most consequential—the policy's two pathways point in opposite directions. Feeding the EwE-simulated fish-population declines through the angler demand function, the same 40% reduction that modestly improves beach conditions cuts predicted angler trips by

[1]A condition where dissolved oxygen levels in the water become too low to support most aquatic life, often creating "dead zones."

30.5%, a welfare loss of roughly $1,048 per angler per season (sensitivity range $349–$1,423), transmitted through a trophic cascade.[2] Per person, the angler's loss is 2 to 12 times the average beachgoer's gain. Nutrient policy, in short, creates identifiable winners and losers across Lake Erie's recreational economy; aggregate net welfare is plausibly positive given the far larger beachgoing population, but the concentrated losses argue for pairing load reduction with fisheries-side measures.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the two surveys, the resulting samples, and the ecological inputs. Section 4 develops the joint RP/SP negative binomial framework and the welfare calculations. Section 5 reports the estimation results, Section 6 the welfare analysis across basins and user groups, and Section 7 concludes.

2 Literature Review

Two literatures frame this study. The first documents Lake Erie's ecological conditions—the blooms, hypoxia, and fish-community dynamics that define what recreators encounter—and the emerging evidence of a water quality–fisheries tradeoff. The second develops the recreation-demand methods we build on: the travel cost model and the joint estimation of revealed and stated preference data. We review each in turn and position our contribution within them.

2.1 Lake Erie's Ecological Setting and the Water Quality–Fisheries Tradeoff

In the summer of 2023, the harmful algal bloom in the western basin of Lake Erie was characterized by NOAA as moderately severe, with a final severity index (SI) of 5.3 (NOAA National Centers for Coastal Ocean Science, 2023a)—within the Severe band of the categorical scale used in our design. This was smaller than the prior year's bloom (SI 6.8). The reduced extent in 2023 was attributed to unusually low precipitation levels in May, which resulted in lower phosphorus runoff into the lake—a key driver of these blooms; the bloom began in early July, fully developed by mid-July, peaked in mid-August, and declined through September, with cooler, windier weather reducing biomass (NOAA National Centers for Coastal Ocean Science, 2023b). Despite its moderate size, the bloom still posed risks to recreation, tourism, and water quality, particularly impacting beach access and drinking water safety for surrounding communities. The central basin of Lake Erie did not experience significant surface HABs comparable to the WB. The central basin typically deals more with hypoxia (low oxygen levels in bottom waters) rather than extensive surface algal blooms, as the water is deeper and less nutrient-rich near the surface compared to the WB. In the

[2] A chain of ecological effects that occurs when a change at one level of the food web causes impacts at multiple lower levels.

CB, hypoxia remained a significant issue, typically beginning in late spring or early summer due to water stratification (NOAA Great Lakes Environmental Research Laboratory, 2023; NOAA National Centers for Coastal Ocean Science, 2024b) which are particularly detrimental to aquatic life and can complicate water treatment processes. The hypoxic zone primarily develops in deeper parts of the basin and progresses from nearshore to offshore. Weather patterns, including strong winds, can cause upwelling, which brings hypoxic water to the surface, potentially affecting water quality at intake sites for drinking water plants. These events require rapid adjustments in water treatment to mitigate effects such as unpleasant taste, odor, and corrosion risks in distribution systems. The eastern basin, being the deepest part of the lake (mean depth of 24 meters and a maximum depth of 63 meters), generally experiences less frequent and less severe hypoxia compared to the central and western basins. However, nutrient inputs and environmental changes still impact its ecosystem. Excessive nutrient loads often lead to nuisance macroalgae (such as *Cladophora*), which can clog water intakes and foul beaches, impacting water quality and ecosystems (U.S. Environmental Protection Agency, 2025; Depew et al., 2022; McCusker et al., 2023).

A parallel ecological literature documents that nutrient abatement itself carries costs for the fishery. The link between nutrient enrichment and fish production is long established (Ney, 1996; Ludsin et al., 2001), and the International Joint Commission has warned that further phosphorus reductions could depress Lake Erie's offshore productivity (International Joint Commission, 2020). Most directly, Scavia et al. (2024) combine a century of Lake Erie commercial-harvest records (1928–2022) with hypoxia modeling and conclude that the current nutrient-load targets appear overly restrictive relative to present fishery needs—clear evidence of a water quality–fisheries tradeoff. That literature quantifies the ecological and commercial-harvest dimensions of the tradeoff; its recreational-welfare dimension, which we estimate here, has not previously been valued.

In summer 2024, the WB bloom was again moderately severe, with a final SI of 6.6—more intense than 2023. The bloom initiated along the Michigan coast in late June, was fully established by early July, peaked in mid-August, and then weakened through September; cooler temperatures and stronger winds in late September accelerated biomass decline. As in 2023, the bloom largely hugged the U.S. shoreline from Michigan to Ohio. Notably, 2024 marked the earliest bloom start in the satellite record since 2002 (July 1). The SI outcome was slightly above the seasonal forecast range (4.5–6) (NOAA National Centers for Coastal Ocean Science, 2024a). In the CB, seasonal hypoxia again developed with summer stratification, and intermittent upwelling events posed operational challenges for nearshore drinking-water intakes (NOAA National Centers for Coastal Ocean Science, 2024b). The EB remained deepest and comparatively less prone to extensive surface cyanobacterial blooms, though

nuisance *Cladophora* growth can occur locally (U.S. Environmental Protection Agency, 2025; Depew et al., 2022). These observed conditions—a final severity index of 5.3 in 2023 and 6.6 in 2024, both within the Severe band—define the revealed-preference baselines used in Section 4.

Lake Erie's three basins differ sharply in depth and thermal regime: the shallow western basin warms earliest, the central basin is intermediate and stratifies seasonally, and the deep eastern basin maintains the most persistent cold-water habitat. These gradients structure species distributions and angler encounters across basins (New York State Department of Environmental Conservation, n.d.; Schertzer, 1987).

Cold-water and cool-water species (e.g., lake trout, steelhead/rainbow trout...) are most associated with deeper, cooler, offshore areas, especially in the eastern basin. Recent binational assessments track these stocks and emphasize the offshore cold-water community's status and management (Great Lakes Fishery Commission, Coldwater Task Group, 2025; New York State Department of Environmental Conservation, Lake Erie Fisheries Research Unit, 2025).

Walleye, yellow perch, and smallmouth bass dominate recreational effort, with strong presence in the western basin and seasonal shifts into central/eastern waters as conditions change. Seasonal changes in temperature and forage how anglers' fish (Ohio Department of Natural Resources, Division of Wildlife, 2021; Ohio Department of Natural Resources, n.d.). Management plans and task-group reports document stock trends and harvest policies for these percids (Great Lakes Fishery Commission, Lake Erie Committee, 2024; Great Lakes Fishery Commission, Yellow Perch Task Group, 2025; Ohio Department of Natural Resources, n.d.).

These ecology patterns motivate our covariate blocks: bloom/odor/bacteria attributes (A_{it}) are expected to bite more in nearshore warm-water fisheries, while catch-rate attributes (F_{it}) are shaped by basin-specific habitat and angling technology that access cold-water strata (Great Lakes Fishery Commission, Lake Erie Committee, 2024; New York State Department of Environmental Conservation, n.d.).

2.2 Recreation Demand Methods and Lake Erie Applications

Many studies have explored the recreational behavior of people taking day trips to the Lake Erie area and its beaches mostly in the western basin due to the higher HAB severity. However, no comprehensive analysis has yet examined recreational activity across all beaches along Lake Erie while simultaneously considering all three basins (eastern, central, and western). For instance, Sohngen et al. (1999) analyzed the value of day trips to two Lake Erie beaches using the travel cost method, focusing on specific locations rather than the lake

as a whole. This study addresses the need for a broader perspective by evaluating all of Lake Erie's beaches within a unified framework.

For the first part of our data, we focus on a survey targeted at beachgoers to uncover the recreational demand function for this group using the travel cost method for three basins of Lake Erie. We build on the work of Wolf et al. (2019), who examined the impacts of harmful algal blooms (HABs) and *E. coli* contamination—a fecal-indicator bacterium used to judge freshwater recreation safety—on recreational behavior in Lake Erie. In their study, they focused on 18 Ohio counties along the Lake Erie shoreline and collected data on trips to the lake during the summer of 2016. Their analysis concentrated on the western basin of Lake Erie, which has been repeatedly exposed to HABs for nearly 20 years. They utilized survey-based approaches and estimations based on the conditional logit, mixed logit, and latent class models of recreation choice. Their findings showed that travel costs had a negative and significant effect on recreational choices, with a value of -0.049. Moreover, they found that water quality measures, particularly algae and *E. coli* concentrations, significantly impacted recreational decisions. The coefficient on algae was negative and significant (-0.0121), confirming the hypothesis that HABs influence visitors' decisions to visit Lake Erie. Similarly, the concentration of *E. coli* had a significant negative impact (-0.0620) on site choice. As an extension of this study, we broaden the analysis to include all three basins of Lake Erie (western, central and eastern) to allow for a more comprehensive comparison of algae bloom impacts. This expansion includes data from Michigan, Pennsylvania, and New York, in addition to Ohio counties facilitating a more regionally representative understanding of how HABs affect recreational behavior across the entire lake.

Lake Erie's harmful algal blooms have also generated a rapidly growing economics literature since Wolf et al. (2019). On the recreation margin, Boudreaux et al. (2023) use choice experiments to measure beachgoer preferences for avoiding blooms and bacterial warnings on Lake Erie beaches, and Wu et al. (2024) estimate recreational angling losses attributable to Lake Erie HABs using revealed-preference count-data models, complementing the discrete-choice evidence of Zhang and Sohngen (2018). Beyond recreation, regional HAB damages have been documented through housing markets—Wolf et al. (2022) find that algae contamination lowers Lake Erie housing prices, Zhang et al. (2022) link satellite-detected cyanobacterial blooms to inland-lake property values, and Georgic et al. (2025) show blooms alter housing renovation decisions—and through defensive behavior, with Liu and Klaiber (2023) estimating household averting expenditure. Our study differs from this literature in jointly estimating RP and SP responses for two user groups on the same lake, in treating bloom extent as a categorical severity factor tied to NOAA classifications, and in linking the estimated demand functions to ecosystem-model simulations of nutrient policy.

The contingent valuation debate that followed the 1989 Exxon Valdez oil spill placed the credibility of stated preference methods under intense scrutiny (Portney, 1994; Carson et al., 2003). One response was to combine stated preference data with revealed preference data, grounding hypothetical responses in observed behavior: Cameron (1992) first joined contingent valuation and travel cost data in a single likelihood, and subsequent work extended the approach to contingent behavior and trip data (Adamowicz et al., 1994; Cameron et al., 1996; Englin and Cameron, 1996). Combining the two data types improves the efficiency of benefit estimation and permits consistency tests between them (Huang et al., 1997). Whitehead et al. (2000) show how stated preference methods can value quality changes at a single site without assuming individuals respond to objective quality measures, and Whitehead et al. (2008), reviewing the state of the science, argue that pooling addresses the central weakness of each source: SP data suffer from hypothetical bias, while RP data alone cannot identify conditions outside observed variation. Given these advantages, we combine RP and SP data in our survey design.

The Travel Cost Method (TCM) is commonly used in environmental valuation literature to estimate recreation demand as a function of trip costs. Travel cost recreation demand models stem from Hotelling's (1947) observation that an outdoor recreation site's services require the consumer to incur the costs of a trip to that site. Travel costs serve as implicit prices meaning that the expenses of reaching a site—transportation, time, and related outlays—act as the price of the trip even though no explicit price is paid in a market. These costs reflect both people's distances from recreation sites and their opportunity costs of time (Whitehead et al., 2000). Parsons (2003) focuses on using observed travel behavior to infer the value individuals place on recreational experiences. He applies the single-site travel cost model, which treats the number of trips taken to a recreational site as a function of travel costs, individual preferences, and site characteristics. Following the approach outlined by Parsons (2003) in the travel cost method, we aim to estimate the recreation demand function for the three basins of Lake Erie.

A methodological choice in this paper is the joint estimation of RP and SP data. Cameron (1992) first combined contingent valuation and travel cost data in a single likelihood, showing that the two data types can identify one underlying preference structure while improving efficiency. Adamowicz et al. (1994) and Adamowicz et al. (1997) formalized the fusion of revealed and stated choices, allowing the data sources to differ in error variance, and Swait and Louviere (1993) established the role of the scale parameter when comparing preference parameters across data sources; in count-data settings this data-source difference is captured by a data-type indicator. Recent applications confirm the value of fusion in recreation contexts: Hindsley et al. (2023) jointly estimate angler revealed site choices and stated choice

experiments while accounting for attribute non-attendance, Hindsley et al. (2022) combine RP and SP models for artificial-reef siting in the Florida Keys, Coote et al. (2021) separate generalizable from source-specific preference heterogeneity in RP–SP fusion, Helveston et al. (2018) address endogeneity that can arise when pooling RP with SP data, and Whitehead and Lew (2020) demonstrate joint estimation of revealed and stated preference discrete choice data for recreation benefits. We follow this literature by stacking each respondent's RP observation with their SP scenario responses and estimating a single negative binomial demand function with a respondent-level random intercept and an SP indicator that absorbs hypothetical bias.

3 Data

Estimating how recreation responds to changes in Lake Erie's ecological conditions requires data that link individual trip behavior to the conditions recreationists encounter. Our outcome of interest is the number of day trips individuals take to the Lake Erie region for beach recreation and recreational fishing, which we model as count data. To measure this behavior, we designed and administered two surveys—one of beachgoers and one of anglers—each eliciting both revealed behavior (trips respondents actually took) and stated behavior (trips they report they would take under hypothetical changes in conditions such as water clarity and fish catch). Figures 1 and 2 map the spatial distribution of survey respondents across the study area, using county, city/township, and state boundary layers we compiled in QGIS. We pair the survey responses with ecological data on water clarity and fish populations. Together, these data allow us to estimate recreation demand as a function of ecosystem characteristics, providing the empirical linkage between the ecological and economic systems that the welfare analysis in Section 6 relies on.

3.1 Survey Type

The beachgoer survey targeted individuals who visit Lake Erie and its shorefront for activities such as sunbathing, swimming, walking pets, or reading; the angler survey targeted those who fish in Lake Erie. Fielding separate instruments allowed us to sample each user group directly; because some individuals both visit beaches and fish, each survey also asked respondents about their participation in the other activity, allowing us to identify and account for this overlap. Both instruments build on an established literature that combines revealed preference (RP) and stated preference (SP) data within the travel cost method to measure demand for beach recreation and recreational fishing (e.g., Huang et al., 1997; Whitehead et al., 2000; Jeon et al., 2011; Huang et al., 2012; Melstrom and Lupi, 2013; Ready et al., 2018; Hunt et al., 2021). The full beachgoer survey appears in Appendix A and the angler survey in

Appendix B; we reference survey items as A# and B#, respectively. Table 1 summarizes the shared and group-specific content of the two instruments and the question number(s) where each item appears.

The RP sections are similar across the two surveys. Both asked respondents to report the number of trips taken to specific Lake Erie locations (following Jeon et al., 2011), trips to destinations other than Lake Erie, home ZIP code, trip expenditures, and demographic characteristics including pre-tax household income, gender, year of birth, marital status, and employment status. Because respondents identified the specific park or shoreline they visited, the trip data support analysis at both the aggregate and basin levels. Both surveys also asked whether respondents had observed algal growth along the shore or in the water, whether they were aware of water-quality and safety advisories for activities such as swimming and fish consumption, and how blooms and advisories affected their plans. The angler survey additionally collected fishing-specific information, including the species targeted at each location and whether respondents fished from a boat, from shore, or both.

The SP questions presented contingent-behavior scenarios asking how ecological changes would affect the total number of day trips respondents would take to Lake Erie—framed over all Lake Erie destinations rather than any specific site, in contrast to the destination-specific RP questions. Both surveys included the algal-bloom scenarios (following Wolf et al., 2019), which crossed bloom size (1, 4, or 8 miles) with the presence or absence of foul odor and of harmful bacteria, for 12 distinct scenarios; each beachgoer was randomly assigned three of the 12, and each angler two. The angler survey additionally informed respondents about factors that cause catch rates to fluctuate and presented scenarios in which cold-water species populations change by -50%, -25%, -10%, 0, or $+10\%$ and warm-water populations by -50%, -25%, -10%, or 0, excluding the no-change pair—19 scenarios in total, of which each angler was randomly assigned two (following Ready et al., 2018). Follow-up questions asked beachgoers which bloom characteristics matter most for their trip decisions and how they would respond to a bloom at their intended beach (A23–A24), and asked anglers how they would respond to increases or decreases in fish populations (B23–B24).

3.1.1 Beachgoing Survey

We used a Qualtrics panel to procure the beachgoer survey sample, with the required criteria that participants be 18 years or older, reside in U.S. counties that border Lake Erie in Ohio, Michigan, New York, or Pennsylvania, and have visited a Lake Erie beach at least once in 2023, so our surveyed sample provides a representative snapshot of Lake Erie visitors. We collected 2,020 responses from the beachgoer survey. Because the demand models are estimated from these data, we can characterize current patterns of recreation and forecast

Table 1: Main differences/similarities between the two surveys

	Beachgoing Survey	**Angler Survey**
Year of data collection	2023	2024
Demographic questions		
Households' income (A29, B30)	✓	✓
Gender (A26, B27)	✓	✓
Year of birth (A30, B32)	✓	✓
Marital status (A27, B28)	✓	✓
Employment status (A28, B29)	✓	✓
Children under 18 in household (A31)	✓	–
Geographic questions		
Destination beach/county/state (A11–A15, B5–B9)	✓	✓
Home zip code/city/state (A6–A8)	✓	–
Home zip code/city (B2–B3)	–	✓
RP questions		
Number of trips taken to specific Lake Erie locations (A11–A15, B6–B9)	✓	✓
Number of trips taken to destinations other than Lake Erie (A20, B19)	✓	✓
Expenditures (A16, B16)	✓	✓
Observing algae growth along the shore or in the water and its impact (A17, B17)	✓	✓
Awareness of advisories and their impact (A18, B18)	✓	✓
Trip Motivations (A10, B20)	✓	✓
Activity preferences (water vs. shore) (A5)	✓	–
Species targeted at each location (B10–B14)	–	✓
Fishing mode (boat, shore, or both) (B15)	–	✓
Primary species group as a driver of trip decisions (B22)	–	✓
SP questions		
Impact of changes in algal bloom characteristics on total number of DAY trips (12 scenarios; 3 per beachgoer, 2 per angler) (A22, B26)	✓	✓
Importance of algal bloom characteristics for trip decisions (A23)	✓	–
Response to an algal bloom impacting the beach they intended to visit (A24)	✓	–
Impact of changes in population of cold- and warm-water species on total number of DAY trips (19 scenarios; 2 per angler) (B21)	–	✓
Responses to increases in fish species populations (B23)	–	✓
Responses to decreases in fish species populations (B24)	–	✓

Note: RP = revealed preference; SP = stated preference; ✓ = the item appears in that survey; – = it does not. Question numbers refer to the beachgoer (A) and angler (B) questionnaires described in Appendices A and B.

how they change under alternative ecological conditions.

In our dataset collected from the survey, we recorded the total number of day trips made by individuals to Lake Erie in 2023 (questions A11 to A15 of the beachgoer questionnaire), along with the change in the number of trips under hypothetical bloom sizes and characteristics. Additionally, we captured data for trips taken during the peak algae bloom season, from July to September. Algae blooms in Lake Erie typically occur during the summer, particularly between these months, when nutrient-rich conditions and warmer water temperatures foster the growth of harmful algae. Moreover, this period coincides with high recreational activity

due to warmer weather, summer vacations, and holiday weekends. Analyzing data from this period is essential, as it provides the most relevant information on how blooms affect recreational activities such as swimming, fishing, and boating and influence people's behavior and preferences during peak usage times. To determine the average cost of each trip per person in dollars, we collected information on the ZIP code, hometown city, and state of residence as the starting point of each journey, as well as the destination access point, county, and state traveled to. These data allowed us to estimate the weighted average distance traveled by each individual in miles. We then multiplied this distance by the average cost per mile for 2023, sourced from the American Automobile Association (AAA) at 81 cents, which includes the cost of gas, vehicle depreciation, and maintenance.

The geographic distribution of respondents to the beachgoer survey is illustrated in Figure 1. According to the map's legend, the colors represent the range of participants from each county. The darkest shade of green indicates regions with the highest number of respondents from Ohio, followed by Michigan, New York, and Pennsylvania.

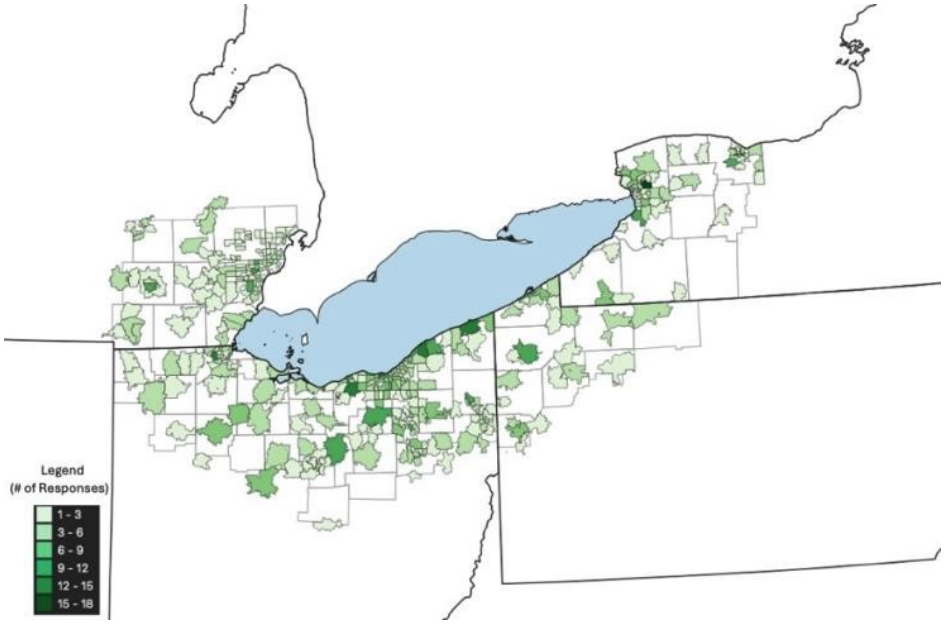


Figure 1: Distribution of participants (recreators)

We applied four sequential data-quality filters to the 2,020 collected responses: (1) respondents reporting zero annual trips were removed, as the sampling frame required at least one 2023 visit; (2) respondents reporting more than 365 day trips per year were removed as implausible; (3) respondents under age 18 were removed; and (4) respondents with missing or non-positive travel cost—cases in which a valid ZIP code or beach destination could not be matched to the distance file—were removed. The final analytic sample contains 1,921 respondents; respondents who visited a Lake Erie beach in 2023 but took no July–September

trips (4% of the sample) are retained with a summer trip count of zero. For basin-level estimation we further define exclusive single-basin visitor subsamples—respondents whose 2023 beach trips fell entirely within one basin (Western Basin $n = 516$; Central Basin $n = 483$; Eastern Basin $n = 423$)—so that each basin model's travel cost unambiguously prices trips to that basin.

Table 2 shows the definitions of the variables and key summary statistics. The beachgoer sample is composed mostly of females (62.6%), with a mean participant age of 44.6 years and mean household income of $63,794. On average, respondents reported 10.21 day trips to Lake Erie beaches between July and September 2023 (SD = 21.36). The mean round-trip travel cost is $71.18 (SD = 64.56), corresponding to a mean one-way distance of 43.94 miles (SD = 39.85); 35% of households include children under 18. As expected, stated trips fall substantially (2.87 day trips) when hypothetical blooms are introduced.

Because the SP questions elicited changes in total Lake Erie trips rather than trips to a specific beach, basin-level estimation requires an assignment rule. We adopt the exclusive single-basin visitor design: each basin model is estimated only on respondents who visited that basin and no other in 2023. Under this restriction the respondent's trips, travel cost, and stated adjustments all refer to the same basin, eliminating the multi-destination measurement error that arises when trips to several basins share one averaged travel cost. The cost of this design is a smaller estimation sample per basin; the benefit is clean identification of basin-specific price and quality responses.

To understand and model motivations for recreational trips, we asked respondents in the beachgoer survey to rate the importance of six beach characteristics: proximity to home, beach size, cleanliness, safety, congestion, and environmental condition. These characteristics were rated using a five-point Likert scale. Safety (89.4%) and cleanliness (83.8%) are rated very or extremely important by the largest shares of respondents, followed by environmental quality (70.0%); proximity to home (34.1%), absence of crowding (39.6%), and beach size (29.1%) matter to far fewer. Cleanliness and environmental condition are thus central to beach choice, consistent with the strong behavioral responses to bloom symptoms estimated below.

Respondents were asked whether they recalled observing algae growth along the shore or in the water during their visits. This was described as "green muck" that might have been unsightly or foul-smelling. Approximately 42% of respondents reported that they did recall such algae growth and 33% reported being aware of water quality advisories at the locations they visited—for example, signage posted at the location (or online) indicating the level of safety related to swimming, general recreation, fish consumption, or water consumption. Of those that experienced a day trip in 2023 where algae growth was present, 48.3% reported no

Table 2: Variable definition and summary statistics for beachgoers

		Overall ($n = 1{,}921$)		**WB** ($n = 516$)		**CB** ($n = 483$)		**EB** ($n = 423$)	
Variable	**Definition**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**
Actual trips (JAS)	July to September day trips to Lake Erie, 2023	10.21	21.36	6.23	14.80	8.25	18.30	8.38	14.10
Expected trips under bloom	Mean stated trips across answered bloom scenarios	2.87	8.94	2.04	5.97	2.41	5.26	2.17	4.53
Travel cost ($)	\$0.81 × weighted one-way distance × 2; basin-specific in basin columns	71.18	64.56	63.97	47.24	52.15	52.29	69.05	71.27
Average distance (mi)	Trip-weighted one-way distance from home	43.94	39.85	39.49	29.16	32.19	32.28	42.63	44.00
Gender	= 1 if female	0.63	—	0.58	—	0.64	—	0.66	—
Age (years)	Respondent's age	44.64	16.51	43.11	16.29	45.55	16.08	45.72	16.85
Household income ($)	Annual household income before taxes	63,794	41,337	65,660	42,223	61,384	39,616	64,836	42,713
Children under 18	= 1 if children under 18 in household	0.35	—	0.36	—	0.33	—	0.34	—
Algae presence	= 1 if observed algae growth during 2023 trips	0.42	—	0.42	—	0.37	—	0.41	—
Advisory presence	= 1 if aware of a water-quality advisory	0.33	—	0.32	—	0.33	—	0.30	—

Note: Statistics computed from the final analytic sample ($n = 1{,}921$, number of visitors). Basin columns are exclusive single-basin visitors (WB $n = 516$; CB $n = 483$; EB $n = 423$) with basin-specific travel cost and distance. Hypothetical trips are each respondent's stated trips across their answered bloom scenarios. Binary variables report shares; SD omitted. \$0.81 = the average cost per mile for 2023, sourced from the American Automobile Association (AAA). SD = standard deviation.

impact on their trip, 26.7% shortened their stay, 32.2% changed the activities they did, and 8.6% went to another beach that was not affected (respondents could select more than one response). These are questions A17 and A18 of the beachgoer questionnaire.

As described in Section 3.1, each beachgoer answered three of the twelve bloom scenarios (a sample question is provided in Appendix A, question A22). On average, the presence of an algae bloom with characteristics such as odor and harmful bacteria led to seven fewer trips per respondent, with 16% reporting no change in their trip behavior. When asked to rank bloom characteristics by importance, respondents placed harmful bacteria first (mean rank 1.72; ranked most important by 50.9%), followed by foul smell (2.38), water quality (2.53), and bloom size (3.36)—an ordering that anticipates the regression results, in which bacteria is the strongest deterrent and bloom extent the weakest (a sample question is provided in Appendix A, question A23).

3.1.2 Angler Survey

The sample for the angler survey is drawn from Lake Erie recreational fishing license holders across the four U.S. states that border Lake Erie. Each state's Department of Natural Resources (DNR) provided us with the email addresses of license holders. We collected 467 responses from the angler survey.

Anglers reported their total day trips to Lake Erie for 2024, since that survey was fielded one year after the beachgoer survey. In addition to recording trip changes under hypothetical bloom sizes and characteristics, the survey also recorded changes in total trips under scenarios that vary cold- and warm-water fish populations, used as a proxy for changes in catch rates. To determine the average cost of each trip per person in dollars, we followed the same procedure as in the beachgoer survey, multiplying the trip-weighted one-way distance by the average cost per mile for 2024, sourced from AAA at 82 cents, which includes the cost of gas, vehicle depreciation, and maintenance.

The geographic distribution of respondents to the angler survey is illustrated in Figure 2.

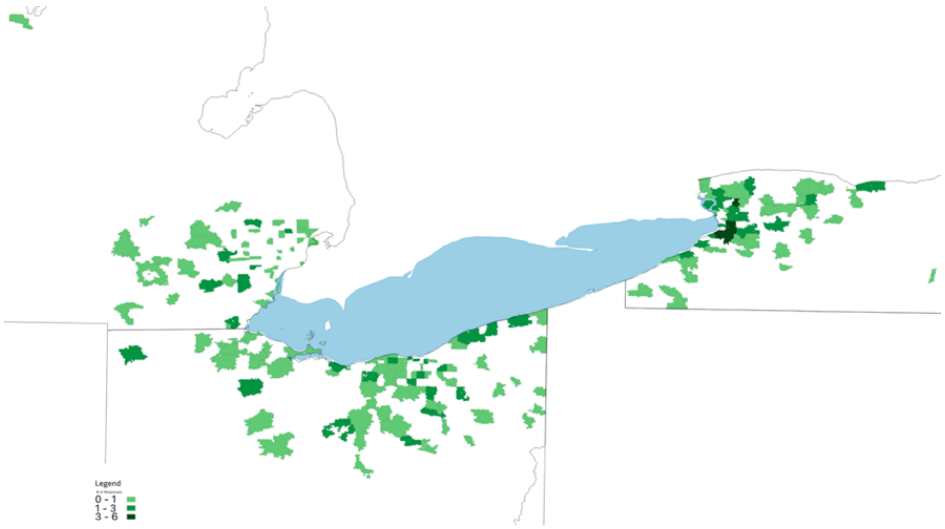


Figure 2: Distribution of participants (Anglers)

We applied analogous filters to the 467 angler responses, removing: (1) non-consenting or preview records; (2) respondents who reported no Lake Erie fishing day trip in 2024 or left the survey before that question; and (3) respondents with zero or missing annual trips, or with more than 365. Excluding these leaves 243 respondents, with 729 observations in each model. Table 3 shows the definitions of the variables and key summary statistics. The angler sample is predominantly male (87.4%), older than the beachgoer sample (mean age 53.4, SD = 16.05), and higher income (mean \$97,251, SD = 41,799); 56.7% own a boat. Respondents reported an average of 17.57 day trips to Lake Erie in 2024 (SD = 26.35). The mean round-trip travel cost is \$67.72 (SD = 61.89), corresponding to a mean one-way distance of 41.29 miles (SD = 37.74). By primary basin fished, the sample comprises 100 Eastern Basin, 88 Western Basin, and 59 Central Basin anglers.

To understand and model motivations for fishing trips, we asked anglers to rate six statements on a five-point Likert scale (1 = not at all accurate, 5 = completely accurate): preferring to fish casually with friends or family; fishing as a favorite hobby; targeting specific

Table 3: Variable definition and summary statistics for anglers

Variable	Definition	Mean	SD
Actual trips (JAS) (2024)	July to September day trips to Lake Erie, 2024	17.57	26.35
Expected trips under bloom scenarios	Mean stated trips across answered bloom scenarios	6.52	11.61
Expected trips under fish-population scenarios	Mean stated trips across the two answered fish-population scenarios	13.89	18.22
Travel cost ($)	\$0.82 × weighted avg one-way distance × 2	67.72	61.89
Average distance (mi)	Trip-weighted one-way distance from home	41.29	37.74
Gender	= 1 if female (sample is 87.4% male)	0.13	—
Age (years)	Respondent's age	53.37	16.05
Household income ($)	Annual pre-tax household income	97,251	41,799
Boat owner	= 1 if respondent owns a boat	0.57	—
Primary basin	EB $n = 100$; WB $n = 88$; CB $n = 59$	—	—

Note: Statistics computed from the final analytic sample ($n = 243$, number of anglers). Primary basin = the basin in which the respondent reported the most 2024 day trips (EB $n = 100$; WB $n = 88$; CB $n = 59$). Hypothetical trips are each respondent's stated trips across their answered bloom and fish-population scenarios. Binary variables report shares; SD omitted. \$0.82 = the average cost per mile for 2024, sourced from the American Automobile Association (AAA). SD = standard deviation.

species; targeting fish by size; caring about the total number caught; and caring about catching at least one fish per trip. Rating each statement as 4 or 5 on the five-point scale, 68% prioritized casual fishing with friends/family, 64% identified fishing as a favorite hobby, 70% reported targeting specific species, 37% emphasized targeting by size, 31% prioritized the total number caught, and 64% emphasized catching at least one fish per trip. These patterns suggest that when choosing a Lake Erie access site for day trips, anglers place greater weight on social context, species targeting, and ensuring at least one catch than on maximizing size or total catch (question B20 of the angler questionnaire).

Respondents were asked whether they recalled observing algae growth along the shore or in the water during their visits—described as green muck that might be unsightly or foul-smelling. Approximately 36% reported observing such algae growth. In addition, 13% reported being aware of water-quality advisories at the locations they visited, such as posted signage or online notices regarding fish consumption, swimming, general recreation, or water consumption.

As described in Section 3.1, each angler answered two of the twelve bloom scenarios (a sample question is provided in Appendix B, question B26). Under these scenarios, stated trips fall to a mean of 6.52 per respondent (SD = 11.61), a decline of about eleven trips from the 17.57 actually taken, with 18% of scenario responses indicating no change in trip behavior. Each angler likewise answered two of the nineteen fish-population scenarios (Appendix B, question B21). Under these scenarios, stated trips fall only to 13.89 (SD = 18.22), a decline of about four trips, with 45% of responses indicating no change, 41% fewer trips, and 14% more. Taken at face value, these figures suggest far greater sensitivity to blooms than to fish populations. The joint models in Section 5 show, however, that most of the bloom-scenario

decline is a uniform downward shift in stated trips across all bloom scenarios—including mild, symptom-free ones—that the SP indicator absorbs, rather than a response to bloom characteristics themselves, whereas the fish-population responses track the scenario-specific population changes with no such shift.

To identify which species groups drive trip decisions under environmental stressors, we asked respondents whether changes in cold-water species, warm-water species, or both equally would more strongly influence their trips. Half (50.6%) chose warm-water species, 39.3% reported both groups equally, and 10.1% selected cold-water species. These responses suggest that shifts in warm-water populations are perceived as the primary driver of trip adjustments for many anglers, although a substantial minority view cold- and warm-water changes as equally important (question B22 of the angler questionnaire).

3.2 Ecological Inputs

The linkage between the nutrient-reduction policy and the demand functions runs through two products supplied by the project's ecological modeling team. It runs in one direction: the simulated ecological conditions enter the demand functions as inputs, and nothing feeds back from recreation to the ecosystem model. The first is an Ecopath with Ecosim (EwE) food-web model of Lake Erie (Christensen and Walters, 2004; Zhang et al., 2023), which simulates *Microcystis* (cyanobacteria) biomass in the Western Basin—where harmful algal blooms originate—under current phosphorus loading and under 40% and 80% load reductions. Simulated biomass is mapped linearly to the bloom severity index (SI) of the NOAA National Centers for Coastal Ocean Science, which classifies bloom conditions as Non-HAB (SI 0–1), Mild (1–3), Moderate (3–5), Severe (5–7), or Very Severe (above 7). Observed conditions place the Western Basin in the Severe category in both survey years (SI = 5.3 in 2023 and 6.6 in 2024); the simulations place it at SI = 4.4 (Moderate) under the 40% reduction and SI = 2.1 (Mild) under the 80% reduction. Severity categories correspond to the bloom extents used in the survey scenarios—Mild to a 1-mile bloom, Moderate to 4 miles, and Severe to 8 miles—so the crosswalk expresses each policy as a change in the bloom conditions respondents evaluated. Table 4, Panel A summarizes the crosswalk.

Because Western Basin blooms are the primary source of cyanobacterial biomass transported into the Central Basin, the ecological team scales Central Basin severity one category below the Western Basin—a Severe Western Basin bloom implies Moderate conditions in the Central Basin, and a Moderate bloom implies Mild—reflecting the downstream transport, dispersion, and attenuation of biomass along the lake's eastward flow. The Eastern Basin is assigned the Mild category throughout: background algae are present in every basin, so a zero-algae state is neither observable nor part of the experimental design.

The second product is the EwE-simulated change in fish biomass under the 40% reduction, by basin and by species group: cold-water biomass falls by 18.2% in the Western Basin, 28.4% in the Central Basin, and 55.8% in the Eastern Basin, and warm-water biomass by 33.0%, 10.9%, and 50.7%, respectively (Table 4, Panel B). Fish-population changes were provided for the 40% reduction only, so the fish-population pathway is evaluated for that policy, with uniform ±10% and ±50% changes for both species groups serving as sensitivity bounds.

Finally, the crosswalk tells us how the policy changes bloom size, but not whether it removes the bloom's symptoms—foul odor and harmful bacteria. The amount of algae in the water is a poor predictor of odor and toxin levels: toxin production varies across algal strains and environmental conditions, most toxins remain inside the cells until the cells break down, and dissolved toxins can linger even after a bloom shrinks (Watson et al., 2008). A smaller bloom is therefore not guaranteed to be a symptom-free bloom. Because we cannot know which case will occur, we evaluate each policy scenario under two bounds: a symptom-free bound, in which odor and bacteria are absent at baseline and under the policy, so the policy changes only the bloom's extent; and a symptomatic-baseline bound, in which today's blooms carry odor and bacteria and the policy removes them wherever it reduces a bloom to Mild. The true welfare effect lies between them. Mild conditions are always treated as symptom-free.

Table 4: Ecological inputs—HAB severity crosswalk and EwE-simulated fish-biomass changes

Panel A: HAB severity crosswalk

Severity index (SI)	Category	Bloom extent	Instances in this study
0–1	Non-HAB	—	—
1–3	Mild	1 mile	80% reduction (SI = 2.1); Eastern Basin baseline
3–5	Moderate	4 miles	40% reduction (SI = 4.4); Central Basin baseline
5–7	Severe	8 miles	Western Basin baseline (SI = 5.3 in 2023; 6.6 in 2024)
> 7	Very Severe	—	—

Panel B: EwE-simulated change in fish biomass under the 40% reduction (%)

Basin	Cold-water	Warm-water
Western Basin	−18.2	−33.0
Central Basin	−28.4	−10.9
Eastern Basin	−55.8	−50.7

Note: Panel A: bloom severity index (SI) of the NOAA National Centers for Coastal Ocean Science, defined for the Western Basin; Central Basin severity is one category below the Western Basin, and the Eastern Basin is Mild throughout. Bloom extents are the scenario levels used in both surveys. Panel B: percentage changes in fish biomass under the 40% phosphorus reduction, simulated with Ecopath with Ecosim by the project's ecological team; changes were provided for the 40% reduction only.

4 Methodology

4.1 Empirical Model

Considering the integer, overdispersed nature of trip counts, we estimate negative binomial count-data models (Winkelmann, 2008); the Poisson model is rejected because its equidispersion restriction is violated in both samples. A further feature of recreation samples is truncation: individuals are observed only if they took at least one trip, which is why the travel-cost literature developed zero-truncated and endogenously stratified estimators (Englin and Shonkwiler, 1995). Neither the angler nor the beachgoer outcome is subject to truncation in this sense, as we explain below. Our central specification is a joint RP/SP negative binomial model in the tradition of Cameron (1992) and Adamowicz et al. (1994): each respondent's actual trip count and their stated trip count under experimentally assigned scenarios are stacked into one dataset and estimated as a single demand function. Joint estimation uses all available information, lets the travel-cost coefficient be identified primarily from real behavior, and lets the environmental-quality coefficients be identified from the controlled scenario variation, while a data-type indicator absorbs the systematic level difference between stated and revealed responses (Swait and Louviere, 1993; Hindsley et al., 2023).

The analysis proceeds in four steps. First, we estimate joint RP/SP negative binomial demand functions for day trips in three families: (i) beachgoer bloom models—a pooled whole-lake model and three basin-specific models on the exclusive single-basin subsamples, each with basin-specific travel cost—which yield each basin's price sensitivity and its response to bloom extent, foul odor, and harmful bacteria; (ii) an angler algal-bloom model with the same attributes; and (iii) an angler fish-population model, which yields the response of angler trips to percentage changes in cold- and warm-water fish populations. Angler models are estimated at the whole-lake level because basin subsamples ($n = 59$–100) are too small to support reliable basin-specific estimation. Second, the travel-cost coefficient of each demand function gives consumer surplus per trip, $\mathrm{CS} = -1/\beta_c$. Third, we translate the 40% (and, for comparison, 80%) phosphorus-reduction policy into the ecological conditions the demand functions take as inputs: the EwE simulations and the HAB severity crosswalk (Section 3.2) determine the bloom severity level in each basin under each policy, and the EwE simulations give basin-specific percentage changes in cold- and warm-water fish populations. Fourth, we predict trips per person under baseline conditions (the 2023 and 2024 bloom conditions actually observed; current fish populations) and under each policy scenario, and value the change in predicted trips at consumer surplus per trip to obtain per-person welfare effects—for beachgoers by basin and for anglers through the bloom and fish-population pathways separately—which Section 6 compares across user groups. Section 5.4 reports

specification and sample robustness checks for the angler models.

The joint dataset is constructed as follows. Each respondent contributes one RP row—their reported trips for the survey year under the environmental conditions actually prevailing in their basin—and one SP row for each hypothetical scenario they answered. For beachgoers this means four rows per respondent: the RP row plus the three algal-bloom scenarios each respondent was assigned. For anglers, the two experiments are analyzed as two separate joint datasets: in the bloom dataset each respondent contributes the RP row plus their two bloom scenarios, and in the fish-population dataset the RP row plus their two fish-population scenarios. A respondent-level random intercept absorbs unobserved heterogeneity in baseline trip frequency, so that the bloom and fish-population coefficients are identified from within-person contrasts between scenarios rather than from cross-person differences in avidity. A stated-preference indicator, $I(\text{SP})$, equal to one for SP rows, captures the systematic level difference between stated and revealed trips under equivalent conditions. Bloom extent enters as a categorical factor with the experimentally assigned levels Mild (1 mile, reference), Moderate (4 miles), and Severe (8 miles); the factor specification permits non-linear responses across severity levels rather than imposing a constant per-mile effect.

Both surveys condition on participation in the survey year—a Lake Erie beach visit in 2023 for beachgoers and a Lake Erie fishing trip in 2024 for anglers—so the estimates describe trip intensity among current recreators, the population whose welfare the policy analysis values, rather than the decision to participate. However, this conditioning does not truncate the dependent variable: the sampling criterion was at least one visit or fishing trip at any time in 2023 or 2024, whereas the modeled outcome is the number of trips taken during July–September. Respondents could have taken no trips during that summer window, so the revealed-preference data contain genuine zeros. Thus, the dependent variable is not zero-truncated, and an untruncated negative binomial model is appropriate for modeling summer trip counts among sampled visitors and anglers.

4.1.1 Joint RP/SP Negative Binomial Specification and Estimation

The negative binomial specification that accommodates overdispersed counts is based on the negative binomial probability distribution:

$$\Pr\left(Y_i = y_i\right) = \frac{\Gamma\left(y_i + \alpha^{-1}\right)}{\Gamma\left(y_i + 1\right)\Gamma\left(\alpha^{-1}\right)} \left(\frac{\alpha^{-1}}{\alpha^{-1} + \mu_i}\right)^{\alpha^{-1}} \left(\frac{\mu_i}{\alpha^{-1} + \mu_i}\right)^{y_i}, \tag{4.1}$$

where y_i is the number of trips for individual i and $\mu_i = \mathbb{E}[Y_i \mid X_i]$ is the expected value or mean of Y_i, which depends on travel cost, individual characteristics, and additional variables such as algal bloom size, foul odor, harmful bacteria, and cold- and warm-water

fish populations. $\alpha > 0$ is the dispersion parameter, which allows for the variance to exceed the mean $\Big(\mathrm{Var}(Y_i \mid X_i) = \mu_i + \alpha\mu_i^2 > \mu_i\Big)$, accounting for overdispersion. $\Gamma(\cdot)$ is the gamma function. We use the log link $\mu_i = \exp(X_i'\beta)$. The moments of the negative binomial distribution are as follows:

$$\mathbb{E}(y_i) = \mu_i, \qquad \mathrm{var}(y_i) = \mu_i + \alpha\mu_i^2.$$

glmmTMB parameterizes the dispersion as $\theta = 1/\alpha$, so that $\mathrm{var}(Y_i \mid X_i) = \mu_i + \mu_i^2/\theta$; Tables 6 and 7 report θ.

Equation (4.2) gives the mean structure of the joint model for the expected number of trips by individual i in observation t:

$$\log[\mu_{it}] = \beta_0 + \beta_c\, C_i + \beta_x'\, X_i + \beta_A'\, A_{it} + \beta_F'\, F_{it} + \beta_{\mathrm{SP}}\, I(\mathrm{SP})_{it} + u_i, \tag{4.2}$$

where $\mu_{it} = \mathbb{E}(y_{it})$ is the expected number of trips of respondent i in row t; C_i is round-trip travel cost; X_i collects individual characteristics (gender, age, children under 18, and income for beachgoers; boat ownership, age, and income for anglers); A_{it} collects algal-bloom attributes (the Moderate and Severe bloom indicators, smell, and harmful bacteria); F_{it} collects the percentage changes in cold- and warm-water fish populations (angler fish-population model only); $I(\mathrm{SP})_{it}$ is the stated-preference indicator; and $u_i \sim N(0, \sigma^2)$ is the respondent random intercept.

Conditional on u_i, the rows $t = 0, 1, \ldots, T_i$ of respondent i (the RP row and the T_i answered scenarios) are independent negative binomial counts with means μ_{it} given by Eq. (4.2), and the marginal likelihood integrates the random intercept out:

$$L(\beta, \sigma^2, \alpha) = \prod_i \int \prod_t p_{\mathrm{NB}}\Big(y_{it}; \mu_{it}(u_i), \alpha\Big)\, \varphi(u_i; 0, \sigma^2)\, du_i, \tag{4.3}$$

where $p_{\mathrm{NB}}(\cdot)$ is the negative binomial pmf above and $\varphi(\cdot)$ the normal density. The likelihood is maximized using the Laplace approximation implemented in glmmTMB (Brooks et al., 2017) with the `nbinom2` parameterization.

For RP rows the environmental attributes must reflect conditions respondents actually faced. We code these from observed bloom severity using the crosswalk described in Section 3.2, which maps the NOAA bloom severity index (SI) for the Western Basin onto the survey's three bloom categories and assigns the Central Basin one category below the Western Basin. For beachgoers (2023), the Western Basin RP row is Severe (SI = 5.3; bloom = Severe, smell = 1, bacteria = 1), the Central Basin Moderate (bloom = Moderate, smell = 1, bacteria = 1), and the Eastern Basin Mild (bloom = Mild, smell = 0, bacteria = 0). The Eastern Basin is

assigned the Mild level rather than a bloom-free state because background algae are present in every basin; a zero-algae lake is not an observable condition and lies outside the experimental design. For anglers (2024) the same crosswalk applies with WB SI = 6.6 (Severe), and each angler's RP row is coded with the conditions of the basin in which they primarily fished. In the fish-population model, RP rows are coded `cold_change` = `warm_change` = 0, defining current fish populations as the baseline against which SP scenarios state percentage changes.

Welfare is computed from the estimated demand function. Consumer surplus per trip is:

$$\mathrm{CS} = -\frac{1}{\beta_c}, \tag{4.4}$$

and the per-person welfare effect of a policy scenario is:

$$\Delta W_i = \left(\mu_i^{\text{policy}} - \mu_i^{\text{baseline}}\right) \times \mathrm{CS}, \tag{4.5}$$

where μ_i^{policy} and μ_i^{baseline} are the model-predicted numbers of trips for individual i under the policy and baseline conditions; Section 6 aggregates these per-person effects, and Appendix C sets out the computation. Because a reduction in bloom biomass does not translate mechanically into a reduction in bloom symptoms—cyanobacterial biomass is an unreliable predictor of taste-and-odor compounds and toxin concentrations (Watson et al., 2008; Section 3.2), each bloom scenario is evaluated under the two bounds defined in Section 3.2: a symptom-free bound (smell = 0, bacteria = 0 at baseline and under the policy) and a symptomatic-baseline bound (smell = 1, bacteria = 1 at baseline, removed where the policy reduces the bloom to Mild); Mild conditions are always symptom-free.

Table 5: Regressors by model and sample

Model	Sample	Regressors
Beachgoer—whole lake	n = 1,921 (7,684 obs.)	travel cost; bloom factor (Moderate, Severe); smell; bacteria; gender; age; children < 18; income/$10k; SP indicator; (1 | respondent)
Beachgoer—basin-specific	Exclusive visitors: WB n = 516; CB n = 483; EB n = 423	basin-specific travel cost; bloom factor; smell; bacteria; gender; age; children < 18; income/$10k; SP; (1 | respondent)
Angler—algal bloom	n = 243 (729 obs.)	travel cost; bloom factor; smell; bacteria; boat owner; age; income/$10k; SP; (1 | respondent)
Angler—fish population	n = 243 (729 obs.)	travel cost; `cold_change`; `warm_change`; boat owner; age; income/$10k; SP; (1 | respondent)
Welfare	All models with a statistically significant travel-cost coefficient	$\mathrm{CS} = -1/\beta_c$; $\Delta W = \Delta\text{trips} \times \mathrm{CS}$; best/worst symptom bounds

Note: RP = revealed preference; SP = stated preference. All models: joint RP/SP negative binomial (glmmTMB, `nbinom2`) with respondent random intercept (1 | respondent) and SP indicator. Bloom extent enters as a factor (Mild = reference). Basin-specific beachgoer models use basin-specific travel cost on exclusive single-basin subsamples. Income enters in $10,000s.

5 Estimation Results

The analysis evaluates how travel costs, demographic factors, algal-bloom characteristics, and fish-population changes shape trip behavior and welfare in Lake Erie and across its Western, Central, and Eastern basins. Because revealed and stated responses are estimated jointly, each coefficient below is identified from both actual behavior and experimentally assigned scenarios. The results reveal substantial spatial heterogeneity, with distinct patterns in trip responses and welfare among the basins.

Tables 6 and 7 present the joint RP/SP negative binomial estimates for beachgoers and anglers, respectively, with standard errors in parentheses. Because the models use a log link, coefficients are interpreted multiplicatively: a one-unit increase in a regressor multiplies expected trips by $\exp(\beta)$. A negative coefficient therefore implies a proportional reduction of $1 - \exp(\beta)$—for example, the Western Basin harmful-bacteria coefficient of -0.432 implies $\exp(-0.432) = 0.65$, a 35% reduction in expected trips—and a positive coefficient a proportional increase (boat ownership, $\beta = 0.963$, implies $\exp(0.963) = 2.6$ times as many trips). The percentage effects reported in the text are obtained in this way.

5.1 Beachgoer Results—Joint RP/SP Negative Binomial Models

Table 6 reports the joint RP/SP estimates for beachgoers: column (1) the pooled whole-lake model (1,921 respondents) and columns (2)–(4) the basin-specific models on exclusive single-basin visitors (WB $n = 516$; CB $n = 483$; EB $n = 423$). Travel cost is negative and precisely estimated in the Western Basin ($\beta = -0.0049$, $p < 0.001$) and Central Basin ($\beta = -0.0057$, $p < 0.001$), implying consumer surplus of \$202.49 and \$174.49 per trip, respectively. In the Eastern Basin the travel-cost coefficient is small and insignificant ($\beta = -0.0008$, $p = 0.35$), so no defensible CS estimate exists for EB; the whole-lake coefficient ($\beta = -0.0010$, $p = 0.034$) is significant but its implied CS ($\approx$ \$996) is inflated by multi-destination averaging in the overall travel-cost measure and is not used for welfare. The SP indicator is large, negative, and highly significant in every beachgoer specification (-1.99 to -1.17), confirming substantial hypothetical bias in stated trip responses and underscoring the value of anchoring the model in revealed behavior.

The bloom-factor coefficients reveal sharply different response structures across basins. In the Western Basin, Moderate ($\beta = -0.170$, $p = 0.017$) and Severe ($\beta = -0.183$, $p = 0.011$) blooms reduce expected trips by a statistically indistinguishable 16–17% relative to Mild conditions—a threshold response in which the presence of a substantial bloom, not its exact extent, deters visitors. In the Central Basin neither Moderate ($p = 0.199$) nor Severe ($p = 0.275$) differs from Mild; CB visitation responds only to smell ($\beta = -0.349$, $p < 0.001$) and bacteria ($\beta = -0.432$, $p < 0.001$). In the Eastern Basin the response is proportional

and monotonic: Moderate reduces expected trips by roughly 20% ($\beta = -0.220$, $p = 0.005$) and Severe by roughly 29% ($\beta = -0.337$, $p < 0.001$)—intuitive for a basin whose visitors are accustomed to bloom-free conditions. Across all basins harmful bacteria is the strongest single deterrent (reducing expected trips by 33–35%), followed by foul odor. Among demographics, trips decline with age everywhere; EB shows additional structure (fewer trips among female respondents, more among households with children under 18). The basin samples are alike in demographics and in how they rank the bloom attributes, so the difference is not one of composition; it is consistent with the conditions each basin's visitors are used to and with how its beaches are used (Central Basin visitors swim least and are the most likely to keep visiting an affected beach; Western Basin visitors have the most water-contact use). Section 7 develops this interpretation.

5.2 Angler Results—Algal Bloom Model

The angler bloom model (Table 7, column 1; 243 respondents) tells a different story. Travel cost is negative and significant ($\beta = -0.0027$, $p = 0.047$), implying CS of \$375.15 per trip—noticeably higher than beachgoer values and consistent with the greater per-trip investment of dedicated anglers. Bloom extent, however, has no detectable effect on fishing trips: relative to Mild conditions, the Moderate coefficient is small, positive, and only marginal ($\beta = 0.181$, $p = 0.089$) and the Severe coefficient is essentially zero ($\beta = -0.005$, $p = 0.959$). Anglers respond to the health-relevant attribute instead: harmful bacteria reduce expected trips by roughly 20% ($\beta = -0.221$, $p = 0.012$), while smell is marginal ($\beta = -0.154$, $p = 0.080$). Boat ownership is the strongest behavioral predictor—boat owners take about 2.6 times as many trips ($\beta = 0.963$, $p < 0.001$)—reflecting both access and avidity. The SP indicator remains large and negative ($\beta = -1.429$, $p < 0.001$), indicating substantial hypothetical bias in the bloom experiment.

5.3 Angler Results—Fish Population Model

The fish-population model (Table 7, column 2; 243 respondents) estimates how trips respond to percentage changes in cold- and warm-water fish populations relative to current levels (RP baseline: `cold_change` = `warm_change` = 0). Travel cost is again negative and significant ($\beta = -0.0029$, $p = 0.020$; CS = \$347.71 per trip—reassuringly close to the bloom-model estimate for the same respondents). Both population variables carry positive, precisely estimated coefficients: a one-percentage-point increase in warm-water populations raises expected trips by 0.72% ($\beta = 0.0072$, $p < 0.001$), roughly twice the cold-water effect ($\beta = 0.0035$, $p = 0.003$)—a pattern consistent with respondents' own assessments in Section 3.1.2, where half identified warm-water species as the group that would most influence their trips. Strikingly, the SP indicator is statistically indistinguishable from zero ($\beta = -0.046$, $p = 0.355$):

Table 6: Joint RP/SP Negative Binomial Models—Beachgoers (2023)

Variable	(1) Whole Lake	(2) Western Basin	(3) Central Basin	(4) Eastern Basin
Panel A: Travel cost and environmental quality attributes				
Travel cost	-0.0010^{*}	-0.0049^{***}	-0.0057^{***}	-0.0008
	(0.0005)	(0.0012)	(0.0011)	(0.0009)
Bloom: Moderate (vs Mild)	-0.0011	-0.1700^{*}	-0.0933	-0.2196^{**}
	(0.0334)	(0.0711)	(0.0727)	(0.0774)
Bloom: Severe (vs Mild)	-0.0921^{**}	-0.1834^{*}	-0.0787	-0.3373^{***}
	(0.0337)	(0.0717)	(0.0721)	(0.0788)
Smell/odor	-0.1596^{***}	-0.1724^{**}	-0.3493^{***}	-0.2657^{***}
	(0.0314)	(0.0655)	(0.0646)	(0.0725)
Harmful bacteria	-0.3183^{***}	-0.4323^{***}	-0.4321^{***}	-0.4098^{***}
	(0.0313)	(0.0655)	(0.0664)	(0.0718)
Panel B: Socioeconomic controls				
Gender (1 = female)	$-0.1170^{\dagger}$	-0.0768	0.0124	-0.3402^{**}
	(0.0627)	(0.1093)	(0.1164)	(0.1309)
Age	-0.0156^{***}	-0.0190^{***}	-0.0197^{***}	-0.0105^{**}
	(0.0019)	(0.0034)	(0.0037)	(0.0039)
Children under 18	0.1728^{**}	0.1194	0.0932	0.5127^{***}
	(0.0668)	(0.1174)	(0.1263)	(0.1371)
Income (per $10,000)	-0.0032	-0.0024	0.0037	-0.0190
	(0.0075)	(0.0132)	(0.0143)	(0.0149)
Panel C: RP/SP correction and constant				
SP indicator	-1.8213^{***}	-1.7890^{***}	-1.9880^{***}	-1.1669^{***}
	(0.0307)	(0.0839)	(0.0847)	(0.0860)
Constant	2.9575^{***}	3.1681^{***}	3.6107^{***}	2.3087^{***}
	(0.1305)	(0.2481)	(0.2679)	(0.2468)
Respondents	1,921	516	483	423
Observations	7,684	2,064	1,932	1,692
AIC	31,663	7,272	7,685	6,515
Random-intercept variance σ_u^2	1.418	1.124	1.177	1.282
Dispersion θ	1.97	2.74	2.24	2.39
CS per trip	\$996[a]	\$202.49	\$174.49	N/A[b]

Notes: glmmTMB, negative binomial (`nbinom2`), respondent random intercept. Dependent variable: July–September day trips (RP rows) and stated trips (SP rows). Bloom extent is a factor with Mild (1-mile) as reference. RP rows coded to 2023 conditions (WB Severe; CB Moderate; EB Mild). Basin-specific travel cost in columns (2)–(4). Standard errors in parentheses. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.10$.[a] Whole-lake CS inflated by multi-destination travel-cost averaging; not used for welfare. [b] EB travel cost insignificant; CS not reported.

unlike the bloom experiments, anglers' stated responses to fish-population scenarios are well calibrated to their actual behavior, lending additional credibility to the welfare estimates built on this model.

Taken together, travel cost is consistently associated with fewer trips, with the largest price sensitivity in the Central Basin and the weakest (statistically insignificant) response in the Eastern Basin. Age reduces trips in every beachgoer model but is insignificant for anglers; gender differences are negligible except in the Eastern Basin, where female respondents take fewer trips; and income is unimportant in every specification.

Table 7: Joint RP/SP Negative Binomial Models—Anglers (2024): (1) Algal Bloom and (2) Fish Population

Variable	(1) Algal Bloom	(2) Fish Population
Panel A: Travel cost, bloom attributes, and fish populations		
Travel cost	-0.0027^{*}	-0.0029^{*}
	(0.0013)	(0.0012)
Bloom: Moderate (vs Mild)	$0.1812^{\dagger}$	—
	(0.1065)	
Bloom: Severe (vs Mild)	-0.0054	—
	(0.1057)	
Smell/odor	$-0.1536^{\dagger}$	—
	(0.0877)	
Harmful bacteria	-0.2205^{*}	—
	(0.0875)	
Cold-water population change (%)	—	0.0035^{**}
		(0.0012)
Warm-water population change (%)	—	0.0072^{***}
		(0.0014)
Panel B: Angler controls		
Boat owner	0.9629^{***}	0.7960^{***}
	(0.1757)	(0.1621)
Age	-0.0059	-0.0040
	(0.0054)	(0.0049)
Income (per $10,000)	0.0125	0.0040
	(0.0203)	(0.0187)
Panel C: RP/SP correction and constant		
SP indicator	-1.4294^{***}	-0.0456
	(0.0824)	(0.0492)
Constant	2.4291^{***}	2.1348^{***}
	(0.3638)	(0.3272)
Respondents	243	243
Observations	729	729
AIC	4,683	4,659
Random-intercept variance σ_u^2	1.293	1.305
Dispersion θ	1.54	11.4
CS per trip	$375.15	$347.71

Notes: glmmTMB, negative binomial (`nbinom2`), respondent random intercept; whole-lake sample (basin subsamples too small for separate estimation). Dependent variable: July–September 2024 day trips (RP rows) and stated trips (SP rows). Bloom extent is a factor (Mild = reference); RP rows coded to 2024 conditions (WB Severe, SI = 6.6; CB Moderate; EB Mild) in column (1) and to `cold_change` = `warm_change` = 0 in column (2). Standard errors in parentheses. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.10$.

5.4 Robustness and Limitations of the Angler Models

The absence of a bloom-extent effect for anglers is a finding, not an artifact of the specification. If bloom size deterred fishing, the Moderate coefficient would be negative and the Severe coefficient more negative still. Instead, Moderate is small and positive ($\beta = 0.181$, $p = 0.089$) and Severe is zero ($\beta = -0.005$, $p = 0.959$)—no dose–response pattern—so we read the marginal Moderate coefficient as sampling noise rather than as evidence that mid-sized blooms attract anglers. A joint Wald test cannot reject that both coefficients are zero

($\chi^2(2) = 4.11$, $p = 0.128$). One caveat applies: with 243 respondents answering two scenarios each, the standard errors on the angler bloom factors (≈ 0.106) are about 40% larger than in the beachgoer basin models (0.071–0.079), so a modest deterrent effect at the Severe level cannot be ruled out. What the data establish firmly is that bacterial contamination—the health-relevant attribute—is the only bloom characteristic that reliably changes angler behavior.

The large SP indicator in the bloom model ($\beta = -1.429$) reflects a level difference between stated and revealed trips, not a distortion of the slope coefficients. The respondent random intercept and the SP indicator absorb person- and mode-specific levels, so the bloom, smell, and bacteria coefficients are identified from within-person contrasts across experimentally assigned scenarios, and the travel-cost coefficient—the welfare denominator—is identified from revealed behavior. The fish-population experiment provides direct evidence that this design can recover well-calibrated responses: its SP indicator is statistically zero ($\beta = -0.046$, $p = 0.355$), indicating that anglers' stated trip adjustments to fish-population changes track their actual behavior.

Two further caveats concern the angler welfare estimates and are addressed in Section 6. First, the consumer surplus per trip implied by the travel-cost coefficients (\$348–\$375) is high relative to the Great Lakes literature; Section 6 shows that repricing the predicted trip loss at literature values of \$75 and \$161 per trip leaves the loss large and negative. Second, 41% of the angler sample fishes primarily in the Eastern Basin, where the EwE simulations project the largest fish declines, so the whole-lake average loss leans toward the most-affected basin; weighting by actual license counts would likely yield a somewhat smaller average loss.

6 Welfare Analysis Across Basins

Ecological inputs. As described in Section 3.2, the HAB severity crosswalk expresses each policy as a change in bloom category by basin—under current conditions the Western Basin is Severe, the Central Basin Moderate, and the Eastern Basin Mild; the 40% reduction moves the Western Basin to Moderate and the Central Basin to Mild, leaving the Eastern Basin at Mild; the 80% reduction moves the Western Basin to Mild as well—and the EwE simulations give the percentage changes in cold- and warm-water fish biomass by basin under the 40% reduction (Table 4). Fish-population changes were provided for the 40% reduction only, so the fish-population pathway is evaluated for that policy, with uniform $\pm 10\%$ and $\pm 50\%$ changes for both species groups as sensitivity bounds.

Scenario construction and assumptions. Each scenario is evaluated by predicting trips per person from the estimated demand functions at the ecological conditions implied by the

policy, comparing them with predictions at baseline conditions, and valuing the difference at consumer surplus per trip (Eqs. (4.4) and (4.5)). Predictions are made for each respondent's revealed-preference row with the random intercept set to its population mean and averaged over respondents. Appendix C sets out the prediction rule and shows how each entry in Table 8 follows from the coefficients in Tables 6 and 7. For beachgoers, each basin model is evaluated at its own basin's bloom category. For anglers, the demand functions are estimated on the whole lake, but the scenario inputs are basin-specific: each angler is assigned the bloom category and the fish-population changes of the basin in which they primarily fish, and the per-angler results are averaged over the sample. Three assumptions deserve emphasis. First, the crosswalk determines the bloom's extent under each policy but not whether its symptoms persist, so each Moderate or Severe bloom scenario is evaluated at two bounds—a best case with no odor or bacteria and a worst case with both—while the Mild category is treated as symptom-free by assumption, on the grounds that background algae at the 1-mile level produce no perceptible odor or bacterial advisory. Second, the bounds apply a consistent symptom assumption to the baseline and policy blooms alike: the worst case treats the observed Western and Central Basin blooms as carrying odor and bacteria (consistent with the RP coding), the best case as symptom-free. Current fish populations define the zero point of the fish-population scenarios. Third, the welfare effects are per person, for current recreators, and are not aggregated to the lake-wide population, for which visitation and license counts would be required.

In this section consumer surplus (CS) is combined with the estimated demand functions to value the ecological changes projected under nutrient-reduction policy. CS per trip equals $-1/\beta_c$: \$202.49 (WB beachgoers), \$174.49 (CB beachgoers), \$375.15 (angler bloom model), and \$347.71 (angler fish-population model); no CS is computed for EB beachgoers. For fish populations, EwE simulations of the 40% reduction imply basin-specific declines of 18.2% (cold) and 33.0% (warm) in the WB, 28.4% and 10.9% in the CB, and 55.8% and 50.7% in the EB, which we apply to each angler according to their primary basin.

For beachgoers, the Western Basin threshold effect dominates the welfare arithmetic. The 40% reduction raises WB trips by only 1.4%, worth \$10.01–\$18.32 per person per July–September season; the 80% reduction, which crosses the threshold into Mild conditions, raises trips 20.1% and is worth \$272.05–\$885.28 per person per season. In the Central Basin the 40% reduction already reaches Mild conditions, so the 40% and 80% scenarios coincide: trips rise 9.8%, worth \$201.02–\$1,315.92 per person per season (the wide bounds reflect whether the baseline Moderate bloom is assumed symptomatic). Eastern Basin conditions are unchanged under every scenario, so EB beachgoer welfare is zero.

For anglers the two pathways point in opposite directions. Through the bloom pathway,

the simulated point estimates are positive—+2.0% trips ($118.22 per angler per season) in the best case and +10.8% ($517.78) in the worst case under the 40% reduction—but these gains rest almost entirely on the bloom-extent coefficients that Section 5.4 shows to be individually and jointly insignificant. The bloom-pathway welfare for anglers is therefore best characterized as not statistically distinguishable from zero, with bacterial contamination the only robust channel. Through the fish-population pathway the policy is costly: applying the basin-specific EwE declines, predicted trips fall from 9.88 to 6.87 per angler per season (−30.5%), a welfare loss of $1,047.60 per angler per season. Uniform-decline sensitivity scenarios bound this between −$348.56 (−10% for both species) and −$1,423.30 (−50% for both species). Because 41% of the angler sample fishes primarily in the Eastern Basin—the basin with the largest simulated fish declines—the whole-lake estimate is weighted toward the most-affected anglers; population-weighted aggregation using license records would likely yield a somewhat smaller average loss.

Our angler CS estimates ($347.71–$375.15 per trip) sit above the typical Great Lakes range—roughly $20–$75 per trip in travel-cost studies and $55–$161 in contingent-valuation studies (2020 dollars)—which is expected for a sample of licensed, actively fishing Lake Erie anglers rather than the general population; Zhang and Sohngen (2018) estimate willingness to pay of $40–$60 per trip for a 40% phosphorus reduction among Lake Erie anglers using a different welfare construct. As a robustness check, repricing the predicted trip loss (−3.01 trips) at literature CS values of $75 and $161 per trip yields annual losses of $226 and $484 per angler—smaller in magnitude but unchanged in sign and economic significance.

Beachgoer and angler welfare are measured for the July–September bloom season, which is when bloom conditions affect beach visits, so the seasonal figure approximates the annual bloom-related gain. The policy therefore creates identifiable winners and losers. Per person, beachgoers in the WB and CB gain $10–$201 per season under the 40% reduction (up to $885 under 80%), while the average angler loses roughly $1,048 per season through the trophic cascade. To put the two groups on the same footing, we also average the basin-specific welfare over the beachgoer sample by primary basin (CB 41%, WB 35%, EB 24%, with EB gaining nothing), just as the angler figure averages basin-specific fish-population changes over anglers: the 40% reduction is worth $89 per beachgoer per season under the symptom-free bounds and $541 if the baseline blooms carry odor and bacteria that the policy removes ($177–$846 under the 80% reduction). Per person, the angler loss is thus 2 to 12 times the beachgoer gain. The aggregate net effect depends on the relative sizes of the two populations, which our surveys do not measure: beachgoers would need to outnumber licensed Lake Erie anglers by roughly 2 to 1 under the symptomatic-baseline bounds, or 12 to 1 under the symptom-free bounds, for their gains to cover the anglers' losses. Beachgoers vastly outnumber licensed

anglers on Lake Erie, so aggregate net welfare is plausibly positive, but the concentrated per-capita losses in the fishing sector are large enough to warrant complementary fisheries policy.

Table 8: Policy-Scenario Welfare Summary—40% and 80% Nutrient Reduction

Scenario	Trip change	Welfare per person
Beachgoers—WB, 40% (Severe→Moderate)	+1.4%	\$10.01 – \$18.32 / season
Beachgoers—WB, 80% (Severe→Mild)	+20.1%	\$272.05 – \$885.28 / season
Beachgoers—CB, 40% = 80% (Moderate→Mild)	+9.8%	\$201.02 – \$1,315.92 / season
Beachgoers—EB, all scenarios (Mild)	0%	\$0
Beachgoers—lake-wide average, 40% (weighted by primary basin)		\$88.52 – \$541.22 / season
Beachgoers—lake-wide average, 80%		\$176.75 – \$845.58 / season
Anglers—bloom pathway, 40% (n.s.‡)	+2.0% to +10.8%	\$118.22 – \$517.78 / season‡
Anglers—fish population, 40% (EwE basin-specific)	−30.5%	−\$1,047.60 / season
Anglers—fish population, sensitivity (−10% both)	−10.1%	−\$348.56 / season
Anglers—fish population, sensitivity (−50% both)	−41.4%	−\$1,423.30 / season

Note: Trip changes are model-predicted percentage changes in mean trips per person relative to the no-reduction baseline; welfare = Δtrips × CS per trip. Beachgoer bloom scenarios are reported as best–worst bounds (symptom-free vs. smell + bacteria); Mild conditions are symptom-free by construction. Angler fish-population welfare uses basin-specific EwE population changes. EB beachgoer welfare is not estimable (travel cost insignificant) and EB conditions are unchanged under all scenarios. ‡ Angler bloom-pathway estimates rest on bloom-extent coefficients that are individually and jointly insignificant (χ^2 (2) = 4.11, $p = 0.128$); this welfare is not statistically distinguishable from zero; Computation detailed in Appendix C. Trip changes are computed holding bloom symptoms unchanged between baseline and policy; in the symptomatic baseline bound, where the policy also removes odor and bacteria, the implied trip increases are correspondingly larger.

7 Conclusions

This study examined both revealed and stated day trips to Lake Erie and how they respond to changes in algal-bloom characteristics—bloom size, foul odor, and harmful bacteria—and to shifts in fish populations used as proxies for catch opportunities, using jointly estimated RP/SP negative binomial demand models linked to Ecopath with Ecosim simulations of the 40% phosphorus-reduction policy.

Among the ecological stressors examined, harmful bacteria exert the strongest deterrent on trip frequency for both user groups, with foul odor a consistent second for beachgoers. Bloom extent matters in a basin-specific way: Eastern Basin beachgoers respond proportionally (larger blooms, fewer trips), Western Basin beachgoers exhibit a threshold response in which Moderate and Severe blooms deter almost identically, and Central Basin beachgoers do not have a statistically significant response to extent. The surveys vary the bloom, not the visitors, so they cannot say why the basins differ. They can rule out composition: the three basin samples are alike in age, income, gender, and children at home (Table 2), and all three rank bacteria as the most important bloom attribute and bloom size as the least (Appendix A, question A23). Two features not tested by the surveys provide plausible reasons for the difference across basins.

The first is visitors' prior experiences. The Eastern Basin rarely has a harmful bloom, so for its visitors any bloom is a departure from clear water and a larger bloom is a larger departure, which is a proportional response. The Western Basin has a Severe bloom nearly every summer, so its visitors already distinguish a real bloom from background algae; what deters them is that a substantial bloom is present, not whether it stretches four miles or eight, which is a threshold. The Central Basin has no large surface bloom of its own; its water-quality problems are hypoxia and the bacterial advisories posted at its urban beaches. For visitors there, the extent of a bloom has little local meaning, while odor and bacteria are exactly the warnings they encounter, and those are the two attributes they respond to.

The second feature is how each basin's beaches are used (Appendix A, questions A5 and A24). Central Basin visitors are the least likely to swim (47%, against 58% in the Eastern Basin and 65% in the Western Basin) and the least likely to boat or kayak; asked what they would do if a bloom reached their beach, they are the most likely to say they would still go and change their activities (39%, against 27% and 31%) and the least likely to move to another beach (25%, against 35–36%). A visitor who walks the shore and would stay regardless has little reason to care how far a bloom extends. Western Basin visitors have the most water-contact use in the sample (65% swim, 33% boat), and for someone entering the water a bloom that reaches the beach is a bloom whatever its size, which is what a threshold looks like. These are explanations the surveys can motivate but not test; a direct test would interact bloom extent with water-contact activity, which we leave for future work.

For anglers, bloom extent is behaviorally irrelevant; only the health-relevant bacteria attribute reduces fishing trips. Managerially, these patterns imply that actions reducing bacterial contamination and odor—and communication that credibly certifies their absence—will generate larger recreational gains than marginal reductions in bloom footprint alone.

Angler behavior responds strongly to fish populations. Warm-water species changes move trips about twice as much as cold-water changes, and linking the demand function to EwE simulations of the 40% phosphorus reduction produces the study's most consequential result: predicted fish-population declines cut angler trips by 30.5%, a welfare loss of roughly $1,048 per angler per season. The bounds around this figure, $349 to $1,423, are not statistical confidence limits; they show how the loss changes with the size of the fish decline, which EwE supplies as point estimates without uncertainty bounds. The lower bound assumes fish populations fall by only 10% in every basin, less than any of the EwE declines, and the upper bound assumes a 50% fall in every basin, close to the largest EwE decline applied lake-wide. Even at the lower bound the loss is about four times the average beachgoer gain under the symptom-free bounds. The same policy that improves beach conditions therefore harms the fishery through a trophic cascade—the recreational-welfare counterpart of the

water quality–fisheries tradeoff that Scavia et al. (2024) document for Lake Erie's commercial harvest, and a result invisible to analyses that study either user group alone.

Taken together, the results show that both water quality and target-species availability are first-order determinants of recreational value in Lake Erie, and that nutrient policy redistributes that value across user groups: Western and Central Basin beachgoers gain \$10–\$201 per person per season under the 40% reduction—and up to \$885 under an 80% reduction, whose fish-population consequences were not simulated, so the 80% comparison covers beach recreation only—while the average angler loses about \$1,048 per season. Averaged over all beachgoers, the gain is \$89 to \$541 per person per season. The width of this range reflects a single unknown: whether a smaller bloom is also a bloom without odor and bacteria. The lower figure assumes the policy changes only the bloom's extent; the upper figure assumes today's blooms carry odor and bacteria and that the policy removes them wherever it brings a bloom down to Mild, which under the 40% reduction is the Central Basin. Because odor and bacteria deter beach trips more than bloom extent does, that assumption matters more than the size of the reduction itself, and the evidence that bloom biomass is a poor predictor of odor and toxin levels (Watson et al., 2008; Section 3.2) makes the lower figure the safer planning number. Taking the angler loss at its central estimate, the per-person angler loss is 2 to 12 times the per-person beachgoer gain, and beachgoers would need to outnumber licensed anglers by between 2 and 12 to one for the policy to break even in aggregate. Converting these per-person estimates into aggregate net welfare therefore requires external counts of the two populations—state fishing-license records and beach-visitation data—a natural extension of this work. Because beachgoers vastly outnumber licensed anglers on Lake Erie, the aggregate net effect is plausibly positive, even though the per-capita losses are concentrated among anglers.

Several limitations qualify these estimates. The angler consumer-surplus values sit above the typical Great Lakes range, though repricing the predicted trip loss at literature values leaves it large and negative; the angler sample leans toward the Eastern Basin, where the simulated fish declines are largest; and the stated bloom responses exhibit substantial hypothetical bias, which the SP indicator absorbs.

Ongoing nutrient-reduction efforts under the GLWQA remain essential, particularly in the Western and Central Basins, and our results argue for pairing them with two complements. First, because beachgoer responses are dominated by bacteria and odor, monitoring and mitigation of bacterial contamination should be prioritized alongside load reduction, and the threshold response implies that ambitious reduction (approaching the 80% scenario for the Western Basin) is required before large beach-recreation gains materialize. Second, because the trophic cascade imposes concentrated losses on anglers, fisheries-side measures—stocking,

habitat restoration, and adaptive harvest management for warm-water species—should accompany nutrient policy to sustain the economic value of the Lake Erie fishery.

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Appendices

Appendix A: Beachgoer Survey Instrument

The beachgoer survey was administered online through a Qualtrics panel in 2023 to residents of the U.S. counties bordering Lake Erie. The questionnaire has 31 items (A1–A31): screening and trip-purpose questions (A1–A5), residence and second-home questions (A6–A9), the importance of beach characteristics (A10), county-by-county visit counts for the access points listed by state (A11–A17), advisory awareness and travel time (A18–A21), the algal-bloom contingent-behavior scenarios (A22), follow-up questions on bloom characteristics and substitution (A23–A25), and demographics (A26–A31). The questionnaire itself does not number its questions; the numbers used in the text and in Table 1 were assigned by the authors in the order in which the questions appear, and the complete questionnaire with this numbering is available from the authors on request.

The contingent-behavior question is reproduced below; each respondent answered it for three randomly assigned scenarios, and the attribute levels are shown with all their possible values separated by slashes. The algal-bloom infographic that accompanied the question (in both surveys) is Figure A1.

A22. Environmental stressors can also contribute to the growth, size, and severity of algal blooms. Algal blooms can produce harmful toxins, be unsightly, and can produce foul-smelling odor. More information on algal blooms is shown in the infographic below, which you can enlarge by clicking on it (the larger image will load in a new window). For the next set of questions, you will be presented with a table that includes characteristics of a hypothetical algal bloom that may occur in the future in Lake Erie. You will be given **three hypothetical scenarios** to consider.

[Infographic: Figure A1.]

Imagine there is an algal bloom that is creating **murky water and a green/brown sludge** and that is impacting a beach or park you intended to visit. The bloom has the characteristics presented in the table below and will last from July to September.

You stated that you took about **...** DAY trips to Lake Erie from July to September in 2023.

	Bloom Characteristics
Distance/Size	Extends 1 mile into the lake/ Extends 4 mile into the lake/ Extends 8 mile into the lake
Smell	No/Yes
Harmful Toxins	No/Yes

About how many total DAY trips would you take from July to September if you encounter an algal bloom with these characteristics?

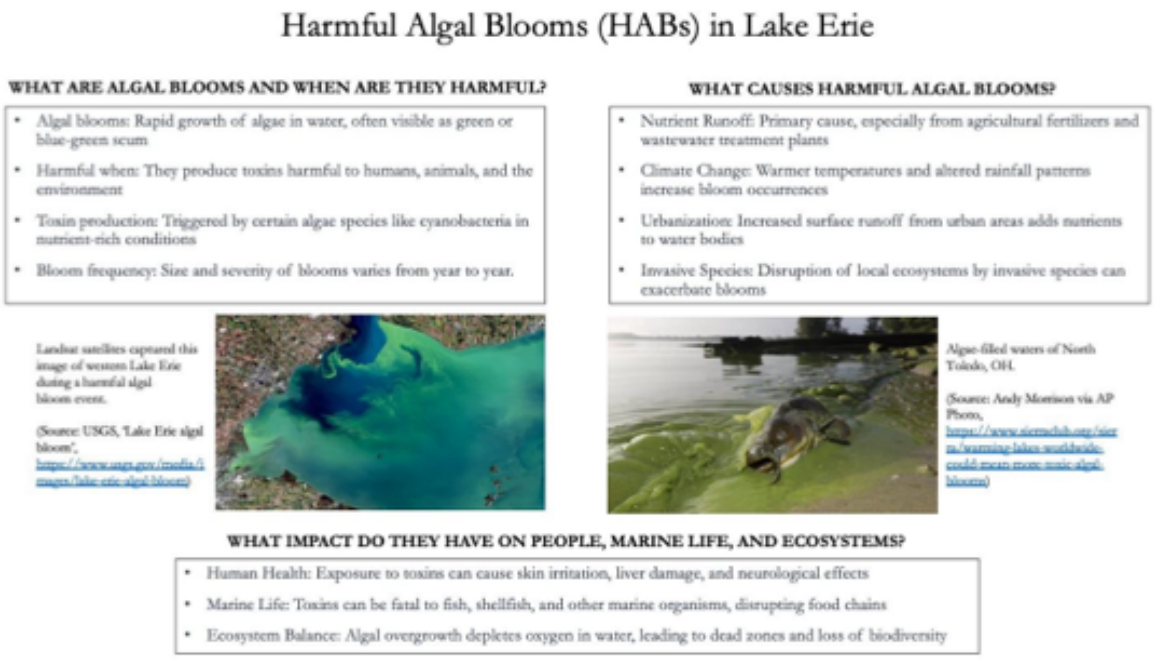


Figure A1: Algal-bloom infographic shown with questions A22 (beachgoer survey) and B26 (angler survey)

Appendix B: Angler Survey Instrument

The angler survey was administered online in 2024 to holders of Lake Erie recreational fishing licenses in Michigan, Ohio, Pennsylvania, and New York, using contact lists provided by each state's Department of Natural Resources. The questionnaire has 32 items (B1–B32): screening and fishing practice questions (B1–B8), fishing locations and trip counts by state (B9–B17), advisory awareness and travel time (B18–B19), attitudes toward catch (B20), the fish-population contingent-behavior scenarios (B21) and their follow-ups (B22–B25), the algal-bloom contingent-behavior scenarios (B26), and demographics (B27–B32). As in Appendix A, the question numbers were assigned by the authors in the order in which the questions appear, and the complete questionnaire with this numbering is available from the authors on request.

The two contingent-behavior questions are reproduced below; each respondent answered each of them for two randomly assigned scenarios, and the attribute levels are shown with all their possible values separated by slashes.

B21. Environmental stressors such as invasive species, algal blooms, and hypoxia can cause the quality of fishing in Lake Erie to change. The next set of questions will present to you **two different hypothetical changes in the population** of cold water (trout and salmon) and warm water (**Smallmouth** bass, etc.) fish species. The percent change in the population of species is associated with a change in the probability of catch.

Shown in the table below is a hypothetical percent change in the population of cold water (trout

and salmon) and warm water species (**Smallmouth** bass, etc.). You may assume that a decrease in population would decrease the probability of catch, whereas an increase in population would increase the probability of catch.

	Fish Population Change
Cold Water	50% Decrease/25% Decrease/10% Decrease/No change/10% Increase
Warm Water	10% Decrease/50% Decrease/25% Decrease/No change/

You stated that you took about **... total DAY trips** to Lake Erie in 2024. About how many total DAY trips would you expect to take if these percent changes occurred in the Lake Erie locations where you fish?

B26. Environmental stressors can also contribute to the growth, size, and severity of algal blooms. Algal blooms can produce harmful toxins, be unsightly, and can produce foul-smelling odor. More information on algal blooms is shown in the infographic below, which you can enlarge by clicking on it (the larger image will load in a new window). you will be presented with a table that includes characteristics of a hypothetical algal bloom that may occur in the future in Lake Erie. You will be given **two hypothetical scenarios** to consider.

[Infographic: Figure A1 in Appendix A.]

Imagine there is an algal bloom that is creating **murky water and a green/brown sludge** and that is impacting a beach or park you intended to visit. The bloom has the characteristics presented in the table below.

You stated that you took about **...** DAY trips to Lake Erie 2024.

	Bloom Characteristics
Distance/Size	Extends 1 mile into the lake/ Extends 4 mile into the lake/ Extends 8 mile into the lake
Smell	No/Yes
Harmful Toxins	No/Yes

About how many total DAY trips would you take if you encounter an algal bloom with these characteristics?

Appendix C: Computing the Policy Scenarios

C.1 Prediction rule

Predicted trips for respondent i under scenario s follow Eq. (4.2) evaluated at the estimated coefficients, with the stated-preference indicator $I(\text{SP})$ set to zero and the random intercept u_i set to its mean of zero:

$$\mu_{is} = \exp\Big(\hat{\beta}_0 + \hat{\beta}_c\, C_i + \hat{\beta}'_x\, X_i + \hat{\beta}'_A\, A_s + \hat{\beta}'_F\, F_s\Big), \tag{C.1}$$

where A_s and F_s are the algal-bloom attributes and the fish-population changes that define the scenario (the bloom models use A_s, the fish-population model F_s). Travel cost C_i and the individual characteristics X_i are held at the respondent's observed values, so only A_s and

F_s change across scenarios. For beachgoers, A_s is the same for every respondent in a basin model; for anglers, it is assigned from the respondent's primary basin. Mean predicted trips under scenario s are $\bar{\mu}_s = \frac{1}{N}\sum_i \mu_{is}$. The per-person welfare effect of Eq. (4.5), averaged over respondents, is $\Delta W = (\bar{\mu}^{\,\text{policy}} - \bar{\mu}^{\,\text{baseline}}) \times \text{CS}$, with $\text{CS} = -1/\hat{\beta}_c$ from Eq. (4.4).

C.2 Percentage changes

Because A_s and F_s enter Eq. (4.2) additively in the log and are common to all respondents in a basin,

$$\frac{\mu_i^{\text{policy}}}{\mu_i^{\text{baseline}}} = \exp\Big(\hat{\beta}'_A \Delta A + \hat{\beta}'_F \Delta F\Big)$$

for every i, where $\Delta A = A^{\text{policy}} - A^{\text{baseline}}$ and $\Delta F = F^{\text{policy}} - F^{\text{baseline}}$. The percentage change in mean trips is therefore a function of the estimated coefficients alone (Table C1). The whole-lake angler change of -30.5% is the average of the three basin changes weighted by each angler's predicted baseline trips (100 EB, 88 WB, and 59 CB anglers in the fish-population model).

Table C1: Percentage change in predicted trips implied by the estimated coefficients

Scenario	**Change in A or F**	$\exp(\hat{\beta}'_A \Delta A + \hat{\beta}'_F \Delta F)$	**Change in trips**
WB beachgoers, 40%, both bounds	Severe → Moderate	$\exp(-0.1700 + 0.1834) = 1.014$	+1.4%
WB beachgoers, 80%, symptom-free	Severe → Mild	$\exp(0.1834) = 1.201$	+20.1%
WB beachgoers, 80%, symptomatic baseline	Severe, 1, 1 → Mild, 0, 0	$\exp(0.1834 + 0.1724 + 0.4323) = 2.199$	+119.9%
CB beachgoers, 40% and 80%, symptom-free	Moderate → Mild	$\exp(0.0933) = 1.098$	+9.8%
CB beachgoers, 40% and 80%, symptomatic baseline	Moderate, 1, 1 → Mild, 0, 0	$\exp(0.0933 + 0.3493 + 0.4321) = 2.398$	+139.8%
WB anglers, fish pathway	cold −18.2, warm −33.0	$\exp(0.003508 \times (-18.2) + 0.007190 \times (-33.0)) = 0.740$	−26.0%
CB anglers, fish pathway	cold −28.4, warm −10.9	$\exp(0.003508 \times (-28.4) + 0.007190 \times (-10.9)) = 0.837$	−16.3%
EB anglers, fish pathway	cold −55.8, warm −50.7	$\exp(0.003508 \times (-55.8) + 0.007190 \times (-50.7)) = 0.571$	−42.9%

Note: Coefficients from Tables 6 and 7; fish-population changes from Table 4, Panel B. Bloom coefficients are relative to Mild, so a change to Mild removes the category coefficient; a symptomatic-to-symptom-free change also removes the odor and bacteria coefficients.

C.3 From trips to dollars

Table C2 reports mean predicted trips at baseline and under the 40% reduction, the change in trips, CS from Eq. (4.4), and ΔW from Eq. (4.5); these are the values summarized in Table 8. Under the 80% reduction, WB beachgoer trips rise to 8.017 on both bounds ($\Delta W = \$272.05$ and $\$885.28$); CB values are unchanged from the 40% case because the basin is already

Mild. The symptomatic-baseline value of $\bar{\mu}^{\text{baseline}}$ (3.645 for the WB, 5.393 for the CB) is the model evaluated at exactly the revealed-preference coding; the symptom-free baseline is the counterfactual with odor and bacteria absent. Because predictions set u_i to zero, they describe a typical respondent with given characteristics rather than the sample mean; multiplying by $\exp(\hat{\sigma}_u^2/2)$ recovers the observed means (for example, $9.881 \times \exp(1.305/2) = 19.0$ against 17.7 reported annual trips among anglers in the model). Percentage changes are unaffected by this choice.

Table C2: Mean predicted trips and welfare per person under the 40% nutrient reduction

Basin or model	**Bound**	$\bar{\mu}^{\text{baseline}}$	$\bar{\mu}^{\text{policy}}$ **(40%)**	**Δ trips**	**CS**	ΔW
WB beachgoers	symptom-free	6.674	6.764	0.090	\$202.49	\$18.32
WB beachgoers	symptomatic baseline	3.645	3.695	0.049	\$202.49	\$10.01
CB beachgoers	symptom-free	11.783	12.935	1.152	\$174.49	\$201.02
CB beachgoers	symptomatic baseline	5.393	12.935	7.542	\$174.49	\$1,315.92
Anglers, fish pathway	—	9.881	6.868	−3.013	\$347.71	−\$1,047.60
Anglers, bloom pathway	symptom-free	15.753	16.068	0.315	\$375.15	\$118.22
Anglers, bloom pathway	symptomatic baseline	12.752	14.132	1.380	\$375.15	\$517.78

Note: Trips are July–September day trips for beachgoers and annual day trips for anglers; ΔW is per person per season (beachgoers) or per year (anglers). Angler bloom-pathway figures rest on bloom-extent coefficients that are individually and jointly insignificant (Section 5.4).